

Lem convergencia uniforme imp ctp

Lema 1. *La convergencia uniforme implica la convergencia puntual.*

Demostración: Sean X, Y espacios métricos y $\forall n \in \mathbb{N} : f_n : X \rightarrow Y$ tales que $f_n \xrightarrow[n \rightarrow \infty]{X} f$

$$\implies \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n \geq N : \forall x \in X : d_Y(f_n(x), f(x)) < \varepsilon.$$

En particular, $\forall x \in X : \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n \geq N : d_Y(f_n(x), f(x)) < \varepsilon$, luego
 $\forall x \in X : f_n(x) \xrightarrow[n \rightarrow \infty]{} f(x).$ ■

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